

Reinforcement Learning for Environment and Agent (MLA0301)

Assessment Tool 2 - Application Based Questions & Solutions

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Course Code: MLA03

Course Name: Reinforcement learning for Environment and Agent

Question 1

Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality.

Solution:

To design an efficient RL framework for ride-sharing driver-passenger matching, we define the core components of the Markov Decision Process (MDP) as follows:

- **State Space (\$S\$):** The state represents the global or localized environment at time \$t\$:
 - Location of available drivers and pending ride requests (GPS coordinates/grid cells).
 - Current time of day, day of the week, and traffic congestion level.
 - Estimated waiting times for unassigned passengers and driver idle times.
- **Action Space (\$A\$):** The action dictates matching or dispatch decisions:
 - Matching a specific driver \$d_i\$ to a passenger \$p_j\$.
 - Repositioning/rebalancing idle drivers to high-demand areas.
 - Declining or deferring a match to wait for better spatial alignment.
- **Reward Function (\$R\$):** Formulated to maximize service efficiency and quality while minimizing delay:
$$R = w_1 * (\text{Trip Fare}) - w_2 * (\text{Passenger Wait Time}) - w_3 * (\text{Driver Pickup Distance/Time}) - w_4 * (\text{Cancellation Penalty})$$
 - A positive reward is given for completed trips based on fare revenue.
 - Negative rewards (penalties) are assigned for higher passenger wait times and long driver pickup distances.
- **Policy & Optimization Goal:** The policy $\pi(a|s)$ learns optimal matching to balance immediate trip fulfillment with long-term driver availability, minimizing overall system waiting times across the network.

Question 3

Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.

Solution:

An RL framework for an autonomous robotic vacuum cleaner focuses on full coverage cleaning, energy efficiency, and obstacle avoidance.

- **State Space (\$\$\$):** Sensor readings (LiDAR/ultrasonic), grid map status (cleaned, uncleaned, obstacle), current battery level, and current position/orientation.
- **Action Space (\$\$):** Discrete directional movements: Move Forward, Turn Left, Turn Right, Reverse, Return to Docking Station.
- **Reward Function (\$R\$):**
 - $+R_{\text{clean}}$ for covering previously uncleaned grid tiles.
 - $-R_{\text{collision}}$ (large penalty) for colliding with obstacles or walls.
 - $-R_{\text{step}}$ (small step penalty) to encourage shortest paths and energy efficiency.
 - $+R_{\text{dock}}$ for returning safely to dock when battery is low.
- **Role of Policy (π):** The policy defines the robot's behavior mapping states to action probabilities. Over time, the policy shifts from random exploration to deterministic navigation along optimal, collision-free cleaning paths.
- **Value Function ($V(s)$ or $Q(s,a)$):** The value function estimates the expected cumulative future reward from a given state/action. As the robot updates its value estimates through temporal-difference learning or dynamic programming:
 - It evaluates long-term outcomes (e.g., avoiding dead-ends early before getting stuck).
 - It prioritizes unexplored high-density dirt areas over already cleaned regions, progressively optimizing total cleaning time and safety.

Question 5

Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

Solution:

A Smart Grid RL framework balances power supply, demand, renewable energy integration, and operational costs in real time.

RL Model Definition:

Component	Description / Elements
State Space (\$\$\$)	Real-time power demand across grid nodes, current generation capacity (solar, wind, thermal), battery storage state of charge (SoC), dynamic electricity market prices, and weather forecasts.
Action Space (\$\$)	Adjusting generator output levels, charging/discharging energy storage systems (batteries), routing power across transmission

Component	Description / Elements
	corridors, and executing load-shedding/demand-response programs.
Reward Function (\$R\$)	$R = -(\text{Operational Cost} + \text{Generation Cost}) - \text{Penalties}(\text{Grid Instability} / \text{Voltage Deviation}) - \text{Penalties}(\text{Unmet Energy Demand})$

Mechanism: The agent receives positive reinforcement for satisfying total load demand while minimizing operating costs and carbon emissions, while incurring steep penalties for grid overload, frequency fluctuations, or rolling blackouts.

Question 6

A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes.

Solution:

Using RL for personalized healthcare dosage adjustment enables continuous dynamic treatment tailored to individual patient responses.

- **State Space (\$S\$):** Patient health indicators including vital signs (blood pressure, heart rate), blood biomarker/drug concentration levels, demographic data, medical history, and side-effect severity index.
- **Action Space (\$A\$):** Dosage adjustment decisions: Increase dosage by Δx , Decrease dosage by Δx , Maintain current dosage, or Pause treatment.
- **Reward Function (\$R\$):**
 - $+R_{\text{health}}$ when target physiological metrics (e.g., blood sugar or blood pressure) move into the optimal therapeutic range.
 - $-R_{\text{side_effect}}$ penalty for induced adverse reactions or toxicity levels.
 - $-R_{\text{ineffective}}$ penalty when biomarker trends show no progress or disease deterioration.
- **How Learning Improves Outcomes:** Patient physiology responds non-linearly to treatments. By updating value estimates based on longitudinal trial data (off-policy or safe offline RL), the RL agent learns personalized treatment strategies that optimize multi-step recovery outcomes rather than short-term symptoms, drastically lowering toxicity risks while achieving stability faster.

Question 10

A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function.

Solution:

An intelligent irrigation framework aims to maximize crop yield while conserving water resources and energy.

- **State Space (\$S\$):**
 - Soil moisture levels at varying root depths.

- Current crop growth stage and water sensitivity.
- Environmental conditions: ambient temperature, humidity, solar radiation, and forecasted rainfall.
- **Action Space (\$AS):** Volumetric water release controls for specific zones: Turn irrigation ON/OFF, Adjust flow rate (Low, Medium, High), and Duration of watering cycle.
- **Reward Function (\$RS):**

$$R = w_1 * (\text{Crop Health} / \text{Growth Index}) - w_2 * (\text{Water Volume Consumed}) - w_3 * (\text{Energy Consumption Cost})$$
- **Optimization Goal:** The system learns to delay unnecessary watering when rainfall is impending and delivers optimal water volume right when soil dryness threatens crop health, maximizing overall water efficiency.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand